DA1001 — Applied Linear Algebra

Tier	A
Phase (order)	1
Do this first	School maths
Estimated effort	12 weeks, ~6 h/week
Unlocks	DA2400, DA3409, DA1002, DA6300

Roadmap

1. Vectors, matrices, Gaussian elimination.
2. Vector spaces, basis, dimension, four fundamental subspaces.
3. Eigenvalues, diagonalisation, symmetric/PSD matrices, Jordan form, SVD.
4. Applications: image processing, networks.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Applied Linear Algebra (48 videos). NPTEL-NOC IITM *Applied Linear Algebra*: IIT Madras, application-oriented, matches DA1001.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- MIT 18.06SC Linear Algebra, Fall 2011 (74 videos). MIT 18.06SC *Linear Algebra* (Strang, Fall 2011): lectures + recitations + problem sets.
- Mathematics for Machine Learning - Linear Algebra (37 videos). Imperial College *Mathematics for ML: Linear Algebra* (official): short, ML-oriented.

⚡ Short videos for intuition and revision

- Essence of linear algebra (16 videos). 3Blue1Brown *Essence of linear algebra*: watch all 16 first, for geometry.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–6	Vector spaces, span, linear independence, basis, dimension, Gaussian elimination. Pair with 3Blue1Brown Essence of Linear Algebra (short videos).
2	7–11	Linear maps, matrices, null space, range, rank-nullity, invertibility.

Week	Lectures	What to do
3	12–18	Solving Ax=b, row operations, quotient spaces, row rank, determinants, change of basis. Implement row reduction to RREF in NumPy.
4	19–23	Polynomials, invariant subspaces, eigenvalues/eigenvectors, diagonalisation, upper-triangular form.
5	24–27	Applications: linear dynamical systems, DFT, counting paths in graphs, PageRank. Implement PageRank by power iteration.
6	28–34	Inner products, Gram-Schmidt, orthogonal complements and projections, least squares, MMSE estimation. Fit a polynomial by least squares.
7	35–42	Adjoints, self-adjoint and normal operators, spectral theorems, positive operators.
8	43–48	Quadratic forms, matrix norms, isometries, SVD, polar decomposition. Compress an image with truncated SVD.

▼ **All lectures in *Applied Linear Algebra* (48 videos)**

1. Applied Linear Algebra _ Course Introduction
2. Vector Spaces: Introduction
3. Linear Combinations and Span
4. Subspaces, Linear Dependence and Independence
5. Basis and Dimension
6. Sums, Direct Sums and Gaussian Elimination
7. Linear Maps and Matrices
8. Null space, Range, Fundamental theorem of linear maps
9. Column space, null space and rank of a matrix
10. Algebraic operations on linear maps
11. Invertible maps, Isomorphism, Operators
12. Solving Linear Equations
13. Elementary Row Operations
14. Translates of a subspace, Quotient Spaces
15. Row space and rank of a matrix
16. Determinants
17. Coordinates and linear maps under a change of basis
18. Simplifying matrices of linear maps by choice of basis
19. Polynomials and Roots
20. Invariant subspaces, Eigenvalues, Eigenvectors
21. More on Eigenvalues, Eigenvectors, Diagonalization
22. Eigenvalues, Eigenvectors and Upper Triangularization
23. Properties of Eigenvalues
24. Linear state space equations and system stability
25. Discrete-time Linear Systems and Discrete Fourier Transforms

26. Sequences and counting paths in graphs
27. PageRank Algorithm
28. Dot product and length in Cn, Inner product and norm in V over F
29. Orthonormal basis and Gram-Schmidt orthogonalisation
30. Linear Functionals, Orthogonal Complements
31. Orthogonal Projection
32. Projection and distance from a subspace
33. Linear equations, Least squares solutions and Linear regression
34. Minimum Mean Squared Error Estimation
35. Adjoint of a linear map
36. Properties of Adjoint of a Linear Map
37. Adjoint of an Operator and Operator-Adjoint Product
38. Self-adjoint Operator
39. Normal Operators
40. Complex Spectral Theorem
41. Real Spectral Theorem
42. Positive Operators
43. Quadratic Forms, Matrix Norms and Optimization
44. Isometries
45. Classification of Operators
46. Singular Values and Vectors of a Linear Map
47. Singular Value Decomposition
48. Polar decomposition and some applications of SVD

Books, notes and websites

- MIT 18.06 (Strang) — OCW; Strang's book (the course text)
- 3Blue1Brown *Essence of Linear Algebra*
- ✅ NPTEL *Linear Algebra* (K.C. Sivakumar, IIT Madras) — nptel.ac.in/courses/111106051
- Book: *Mathematics for Machine Learning* — mml-book.com (free)

Practice / proof of understanding

Do the 18.06 problem sets; implement SVD-based image compression in NumPy.

What your IITM course sheet says

From the PDF: DA1001

Official description: Real-world systems or scenarios throw at us lot of information which needs to be understood and processed. This analysis becomes easier when the available information is organised in a certain structure, which makes it easier to understand the underlying information in a better way. This course on Linear Algebra provides us with mathematical abstractions and structures to organize, analyze and

interpret information and is one of the fundamental tools in any engineering studies. The course will start with an introduction to vector spaces and subspaces, linear functions, systems of linear equations, matrices as linear transformations, matrix algebra, with engineering applications.

Official syllabus (trimmed):

> What is Linear Algebra: Vectors, dot products, linear combinations, linear functions, Matrices. Systems of Linear Equations: Gaussian elimination, Elementary row operations Vector spaces and subspaces: Spaces of vectors, span, basis, linear independence, dimension, rank nullity theorem, four fundamental subspaces, change of basis, complete solution to a system of linear equations, orthonormal bases. Matrices: Eigenvalues and Eigenvectors, Diagonalisation of matrices, Symmetric matrices, Positive definite matrices, generalized Eigenvectors, Jordan form, Singular value decomposition. Linear transformation, matrix representation of a linear transformation. Applications: Image processing, Optimisation, Networks,

Official textbooks: S. Lang, Linear Algebra, 3rd edition, Springer, 2004. Gilbert Strang, Introduction to Linear Algebra, 3rd Edition.

Official references: K. Janich. Linear Algebra. Springer. 1994. H. Anton. C. Rorres. Elementary Linear Algebra: Applications, Wiley; 11th edition, 2013